

Task:

Cybersecurity

Task2(2026) –

future interns

Topic: Phishing

Email Detection &

Awareness system

Table of Contents

Front page.....	1
Table of Contents.....	2
Executive Summary.....	3
Risk Rating.....	4
Tools used.....	5
Phishing email samples analysed.....	6-12
Common Phishing Indicators.....	13
How The Attack Works.....	14
Prevention and Awareness Guidelines.....	15
Conclusion.....	16
References.....	17

Executive summary

Phishing remains as one of the most common and damaging cyber threats to businesses. Attackers impersonate trusted brands to trick users into clicking malicious links or sharing credentials. This report shows and analyses three phishing email samples using header analysis and HTML inspection tools. Each sample shows or rather demonstrates how attackers exploit urgency, fake domains and missing authentication records to deceive recipients.

Risk Rating

Email	Subject	Risk Level	sender	Authentication	Main indicator	Classification
Sample 1	Bradesco Livelos Points	High	Banco.bradesco@ubuntu	SPF/DKIM/DMARC failed	Fake banking domain	Phishing
Sample 2	Microsoft sign-in activity	High	No-reply@access-accsecurity.com	SPF none, DKIM none, DMARC perm Error	Fake Microsoft alert	Phishing
Sample 3	Solar Panel offer	Medium-high	zonnepaneel@apppj.serenitepure.fr	SPF none, DKIM none, DMARC none	Fake solar promotion	Phishing

Tools used

TOOL	PURPOSE
Google message header analyser	Extract and interpret email header metadata
Mx toolbox email header analyser	Verify SPF, DKIM & DMARC records
Text editor(notepad)	View raw HTML content safely

Phishing email samples analysed

Sample 1: Bradesco Livelo Points Expiration

Subject: CLIENTE PRIME – BRADESCO LIBELO:

Seu cartão tem 92.990 pontos LIVELO expirando hoje!

From: banco.bradesco@ubuntu-s-1vcpu-35gb-intel-sfo3-06

SenderIP: 137.184.34.4 (blacklisted)

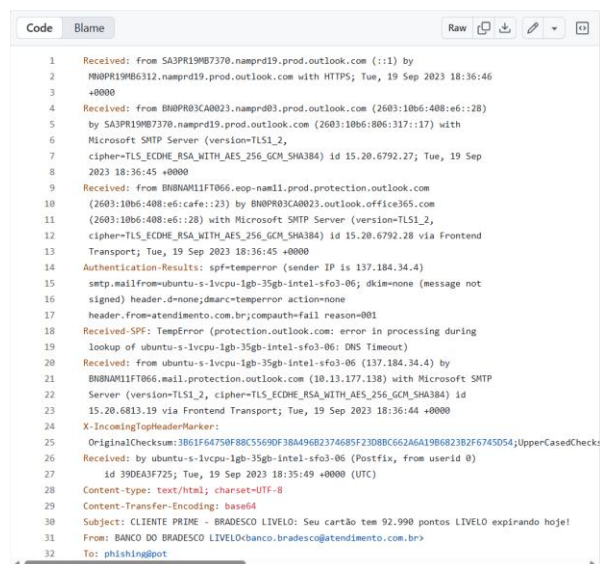
Header Analysis:

- SPF = temperror (IP unknown)
- DKIM = none (message not signed)
- DMARC = temperror
- Authentication failed – spoofed sender

Link Inspection:

The domain blogseguimentmydomain2bra.me is unrelated to Bradesco and currently inactive.

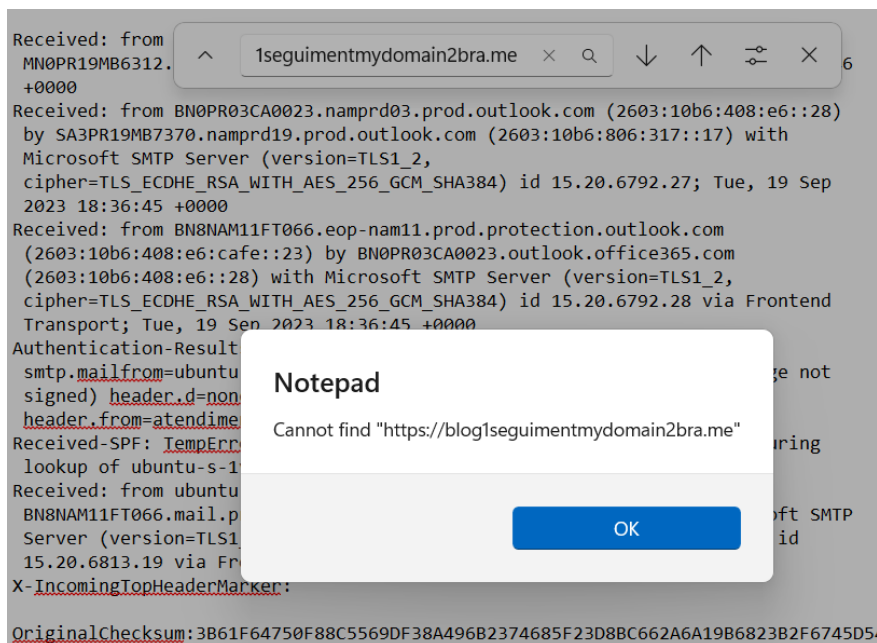
Evidence:



```
1 Received: from SA3PRI9B7370.namprd19.prod.outlook.com (::1) by
2 MB0PR19B6312.namprd19.prod.outlook.com with HTTP5; Tue, 19 Sep 2023 18:36:46
3 +0000
4 Received: from B06PR03CA0023.namprd03.prod.outlook.com (2603:10b6:408:e6::28)
5 by SA3PRI9B7370.namprd19.prod.outlook.com (2603:10b6:806:317::17) with
6 Microsoft SMTP Server (version=TLS1_2,
7 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6792.27; Tue, 19 Sep
8 2023 18:36:45 +0000
9 Received: from B06NAM11FT066.eop-nam11.prod.protection.outlook.com
10 (2603:10b6:408:e6:cafe::223) by B06PR03CA0023.outlook.office365.com
11 (2603:10b6:408:e6::28) with Microsoft SMTP Server (version=TLS1_2,
12 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6792.28 via Frontend
13 Transport; Tue, 19 Sep 2023 18:36:45 +0000
14 Authentication-Results: spf=temperror (sender IP is 137.184.34.4)
15 smtp.mailfrom=ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06; dkim=none (message not
16 signed) header.d=none; dmarc=temperror action=none
17 header.from=atendimento.com.br;compauth=fail reason=001
18 Received-SPF: TempError (protection.outlook.com: error in processing during
19 lookup of ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06: DNS Timeout)
20 Received: from ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 (137.184.34.4) by
21 B06NAM11FT066.mail.protection.outlook.com (10.13.177.138) with Microsoft SMTP
22 Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id
23 15.20.6813.19 via Frontend Transport; Tue, 19 Sep 2023 18:36:44 +0000
24 X-IncomingTopHeaderMarker:
25 OriginalChecksum:3B61F64759F88C5569DF38A496B2374685F23DB8C662A6A19B6B23B2F6745D54;UpperCasedChecksum:
26 Received: by ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06 (Postfix, from userid 0)
27 id 39DEA3F725; Tue, 19 Sep 2023 18:35:49 +0000 (UTC)
28 Content-type: text/html; charset=UTF-8
29 Content-Transfer-Encoding: base64
30 Subject: CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
31 From: BANCO DO BRADESCO LIVELO(banco.bradesco@atendimento.com.br)
32 To: phishing@pot
```

This is the raw email

MessageId	20230919183549.39DEA3F725@ubuntu-s-1vcpu-1gb-35gb-intel-sfo3-06
Created at:	9/19/2023, 8:35:49 PM GMT+2 (Delivered after 56 sec)
From:	BANCO DO BRADESCO LIVELO<banco.bradesco@atendimento.com.br>
To:	phishing@pot
Subject:	CLIENTE PRIME - BRADESCO LIVELO: Seu cartão tem 92.990 pontos LIVELO expirando hoje!
SPF:	temperror with IP Unknown! Learn more
DKIM:	none Learn more
DMARC:	temperror Learn more



Risk Classification: Phishing

Indicators: Fake sender domain, failed authentication, malicious redirect link, urgency language.

Sample2- Microsoft Account Unusual Sign-in Activity

Subject: Microsoft account unusual sign-in activity

From: no-reply@access-accsecurity.com

Return -path: bounce@thcultarfdes.co.uk

SenderIP: 89.144.44.2

Header Analysis:

- SPF = None (sender IP not authorised)
- DKIM = None (email not digitally signed)
- DMARC = PermError (DMARC validation failed because of configuration problems)
- Domain access-accsecurity.com is not a Microsoft domain.

Link inspection: HTML body contains a redirect to sotrecognizd@gmail.com a fake security domain.

Evidence:

```
Code Blame Raw [Icons]
1 Received: from SJOPI19MB6679.namprd19.prod.outlook.com (::1) by
2 MN0PR19MB6312.namprd19.prod.outlook.com with HTTPS; Fri, 8 Sep 2023 05:47:06
3 +0000
4 Received: from DB8P191CA0014.EURP191.PROD.OUTLOOK.COM (2603:10a6:10:130::24)
5 by SJOPI19MB6679.namprd19.prod.outlook.com (2603:10b6:a03:477::19) with
6 Microsoft SMTP Server (version=TLS1_2,
7 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6768.30; Fri, 8 Sep
8 2023 05:47:04 +0000
9 Received: from DB8EUR06FT032.eop-eur06.prod.protection.outlook.com
10 (2603:10a6:10:130:cafe::9b) by DB8P191CA0014.outlook.office365.com
11 (2603:10a6:10:130::24) with Microsoft SMTP Server (version=TLS1_2,
12 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.6768.30 via Frontend
13 Transport; Fri, 8 Sep 2023 05:47:04 +0000
14 Authentication-Results: spf=none (sender IP is 89.144.44.2)
15 smtp.mailfrom=thcultarfdes.co.uk; dkim=none (message not signed)
16 header.d=none;dmARC=permerror action=none header.from=access-accsecurity.com;
17 Received-SPF: None (protection.outlook.com: thcultarfdes.co.uk does not
18 designate permitted sender hosts)
19 Received: from thcultarfdes.co.uk (89.144.44.2) by
20 DB8EUR06FT032.mail.protection.outlook.com (10.233.253.34) with Microsoft SMTP
21 Server id 15.20.6768.30 via Frontend Transport; Fri, 8 Sep 2023 05:47:04
22 +0000
23 X-IncomingTopHeaderMarker:
24 OriginalChecksum:60B22130536EEE635A126CD816D1A4DD037580CD58231D7469764E41AA1094BF;UpperCasedChecksum:
25 From: Microsoft account team <no-reply@access-accsecurity.com>
26 Subject: Microsoft account unusual signin activity
27 To: phishing@pot
28 Content-Length: 18708448
29 Content-Length: 1821750
30 Date: Fri, 8 Sep 2023 05:47:04 +0000
31 Reply-To: sotrecognizd@gmail.com
32 Content-Type: text/html; charset="UTF-8"
```

This is the raw email

Message-ID:	032672b4-77ca-42f8-a036-9711e91bd1f3@DB8EUR06FT032.eop-eur06.prod.protection.outlook.com
Created at:	9/8/2023, 7:47:04 AM GMT+2 (Delivered after)
From:	Microsoft account team _<no-reply@access-accsecurity.com>
To:	phishing@pot
Subject:	Microsoft account unusual signin activity
SPF:	none Learn more
DKIM:	none Learn more
DMARC:	permerror Learn more

Headers Found

Header Name	Header Value
Authentication-Results	spf=none (sender IP is 89.144.44.2) smtp.mailfrom=thcultarfdes.co.uk; dkim=none (message not signed) header.d=none;dmARC=permerror action=none header.from=access-accsecurity.com;
Received-SPF	None (protection.outlook.com: thcultarfdes.co.uk does not designate permitted sender hosts)
X-IncomingTopHeaderMarker	OriginalChecksum:6DB22130536EEE635A126CD816D1A4DDD37580CD58231D7469764E41AA1094BF;UpperCasedChecksum:F9ACD9D1D69FA775F257E981E62B95CD78A6FFF839CE508388DC5167E03028C1;SizeAsReceived:385;Count:12
From	Microsoft account team _<no-reply@access-accsecurity.com>
Subject	Microsoft account unusual signin activity
To	phishing@pot
Content-Length	18708448
Date	Fri, 8 Sep 2023 05:47:04 +0000
Reply-To	sotrecognizd@gmail.com
Content-Type	text/html; charset="UTF-8"
Content-Transfer-Encoding	8bit
X-IncomingHeaderCount	12
Message-ID	<032672b4-77ca-42f8-a036-9711e91bd1f3@DB8EUR06FT032.eop-eur06.prod.protection.outlook.com>
Return-Path	bounce@thcultarfdes.co.uk
X-MS-Exchange-Organization	08 Sep 2023 05:47:04.2927 (UTC)

OriginalChecksum:6DB22130536EEE635A126CD816D1A4DD037580CD58231D7469764E41AA1094BF;UpperCase
dChecksum:F9AC9D1D69FA775F257E981E62B95CD78A6FFF839CE508388DC5167E03028C1;SizeAsReceived:3
85;Count:12
From: Microsoft account team , <no-reply@access-accsecurity.com>
Subject: Microsoft account **unusual** **signin** activity
To: phishing@pot
Content-Length: 18708448
Content-Length: 1821750
Date: Fri, 8 Sep 2023 05:47:04 +0000
Reply-To: sotrecognizd@gmail.com
Content-Type: text/html; charset="UTF-8"
Content-Transfer-Encoding: 8bit
X-IncomingHeaderCount: 12
Message-ID:
<032672b4-77ca-a036-9711e91bd1f3@DB8EUR06FT032.eop-eur06.prod.protection.outlook.com>
Return-Path: bounce@thculturalfdes.co.uk
X-MS-Exchange-Organization-ExpirationStartTime: 08 Sep 2023 05:47:04.2927

It creates Fear and

Urgency

```
<tr style= neight: 18px; >  
<td id="i3" style="padding: 25px 0px 0px; font-size: 14px; font-family: 'Segoe UI', Tahoma,  
/Verdana, Arial, sans-serif; color: #2a2a2a; height: 18px; width: 696px;">We detected  
something unusual about a recent sign-in to the Microsoft account <a id="iAccount"  
:class="link" dir="ltr" style="color: #2672ec; text-decoration: none;"  
href="mailto:sotrecognizd@gmail.com?"  
&cc=sotrecognizd@gmail.com&Subject=Report+The+User">phishing@pot</a>.</td>  
</tr>  
<tr style="height: 18px;">  
<td id="i4" style="padding: 25px 0px 0px; font-family: 'Segoe UI Bold', 'Segoe UI  
Semibold', 'Segoe UI', 'Helvetica Neue Medium', Arial, sans-serif; font-size: 14px; font-  
weight: bold; color: #2a2a2a; height: 18px; width: 696px;"><strong>Sign-in details</strong>  
</td>  
</tr>
```

it uses a mailto: link

instead of taking users to Microsoft's official security portal.

Risk Classification: Phishing

Indicators: spoofed Microsoft sender, no authentication. Fake security domain

Sample 3- Solar Panel Offer

Subject: Zonnepanelen vooreen goede prijs

From: zonnepaneel@appji.serenitepure.fr

SenderIP: 57.128.69.202

Header Analysis:

- SPF = none - the sender could not be verified, increasing the likelihood that email address was spoofed.
- DKIM = none - the sender could not be verified, increasing the likelihood that email address was spoofed.
- DMARC = permerror
- Domain serenitepure.fr unrelated to energy providers.

Link Inspection: No <a href> tags found- plaintext URL embedded in body.

Evidence:

```
Code Blame
Raw [icon] [icon] [icon] [icon] [icon]

1 Received: from MN2PR19MB3966.namprd19.prod.outlook.com (::1) by
2 MN2PR19MB6312.namprd19.prod.outlook.com with HTTPS; Thu, 3 Nov 2022 04:56:17
3 +0000
4 Received: from AM7PR02CA0005.eurprd02.prod.outlook.com (2603:10a6:20b:100::15)
5 by MN2PR19MB3966.namprd19.prod.outlook.com (2603:10b6:208:1e8::10) with
6 Microsoft SMTP Server (version=TLS1_2,
7 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.5769.16; Thu, 3 Nov
8 2022 04:56:16 +0000
9 Received: from VI1EUR05FT048.eop-eur05.prod.protection.outlook.com
10 (2603:10a6:20b:100::4) by AM7PR02CA0005.outlook.office365.com
11 (2603:10a6:20b:100::15) with Microsoft SMTP Server (version=TLS1_2,
12 cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.5791.22 via Frontend
13 Transport; Thu, 3 Nov 2022 04:56:15 +0000
14 Authentication-Results: spf=none (sender IP is 57.128.69.202)
15 smtp.mailfrom=dturm.de; dkim=none (message not signed)
16 header.d=none; dmarc=none action=none
17 header.from=appjj.serenitepure.fr; compauth=fail reason=001
18 Received-SPF: None (protection.outlook.com: dturm.de does not designate
19 permitted sender hosts)
20 Received: from dturm.de (57.128.69.202) by
21 VI1EUR05FT048.mail.protection.outlook.com (10.233.243.188) with Microsoft
22 SMTP Server id 15.20.5791.20 via Frontend Transport; Thu, 3 Nov 2022 04:56:15
23 +0000
24 X-IncomingTopHeaderMarker:
25 OriginalChecksum:9F3096EACB003DF9CDCAFE3B2131300B1E3B982DA325FC1E8548F8F8B2C2F79;UpperCasedChecks
26 Subject: =?UTF-8?B?8J+Ujw==?= Zonnepanelen voor een goede prijs
27 From: "Zonnepanelen installateur" <zonnepaneel@appjj.serenitepure.fr>
28 Sender: "Zonnepanelen installateur" <zonnepaneel@appjj.serenitepure.fr>
29 Reply-To: "Zonnepanelen installateur" <news@aichakandisha.com>
30 To: phishing@pot
31 In-Reply-To: _-...-xf98avdmffvo8ke75+wtbdr+1+++_er/zw-81-7cpfrbgjnl-142022132846@rxgmmq5n0a1.fichi-4
32 List-Unsubscribe-Post: List-Unsubscribe=One-Click
33 X-Banner: Abuse: abuse@protection.outlook.fr
```

This is the raw email

Message-ID	0.0.0.0.1D8EF409A5C12CE.37AA@dturm.de
Created at:	11/3/2022, 6:56:15 AM GMT+2 (Delivered after)
From:	"Zonnepanelen installateur" <zonnepaneel@appjj.serenitepure.fr>
To:	phishing@pot
Subject:	 Zonnepanelen voor een goede prijs
SPF:	none Learn more
DKIM:	none Learn more
DMARC:	none Learn more

SPF and DKIM Information

dmARC:access-accsecurity.com Hide Solve Email Delivery Problems

Status	
Name	DMARC Record Published
Response	No DMARC Record found

 [More Info](#)

Reported by **e.gtld-servers.net** on 6/27/2026 at **10:29:04 AM (UTC 0)**, [just for you.](#) [Transcript](#)

spf:thculturalfdes.co.uk:::1 Show Solve Email Delivery Problems

Dkim Signature Error:
No DKIM-Signature header found - [more info](#)

Dkim Signature Error:
There must be at least one aligned DKIM-Signature for the message to be considered aligned.
- [more info](#)

Risk Classification: Phishing

Indicators: unverified sender, generic offer, no authentication records.

Common Phishing Indicators

Indicator	Description
Fake sender domain	Domain does not match organization name
Failed SPF/DKIM/DMARC	Email cannot be verified as legitimate
Urgency or fear language	“Expiring today”. “Unusual activity”
Suspicious links	Redirects to non-official domains
Generic Greetings	“Dear user “instead of personal name

How The Attack Works

Attackers create emails that look real/legit, but it redirects users to fake website so that they can steal credentials or financial data. They usually use spoofed domains and disable authentication records to bypass filters.

Prevention and Awareness Guidelines

✓ DO'S

- Verify sender domain before clicking links
- Hover over links to see real destination
- Report suspicious emails to IT or security team
- Use multi-factor authentication for accounts

⊘ Don'ts

- Don't click unknown links or attachments
- Don't share password or OTP codes via email
- Don't ignore authentication warnings from email clients

Conclusion

All the three analysed emails show common phishing techniques used by cybercriminals, including domain spoofing, missing email authentication records, misleading branding, fake sender identities and urgency-based social engineering. The header analysis together with inspection of sender domains and embedded links clearly shows each email as malicious and avoiding suspicious links is an effective method for detecting phishing attempts before users become victims.

References

- MxToolbox. (n.d.). *Email Header Analyzer, RFC822 parser*.
<https://mxtoolbox.com/Public/Tools/EmailHeaders.aspx>